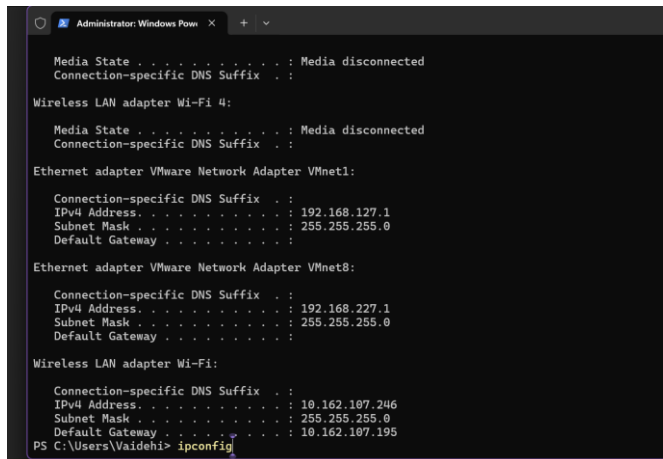


Troubleshooting Tools

Q.1 Use ipconfig to find the IPv4 address and default gateway

Solution: Find IPv4 Address and Default Gateway

I used the **ipconfig** command in Windows Command Prompt to find the network configuration of my computer.

A screenshot of a Windows Command Prompt window titled 'Administrator: Windows PowerShell'. The window shows the output of the 'ipconfig' command. It displays network configuration for three adapters: 'Wireless LAN adapter Wi-Fi 4:', 'Ethernet adapter VMware Network Adapter VMnet1:', and 'Ethernet adapter VMware Network Adapter VMnet8:'. The 'Wi-Fi 4' adapter shows an IPv4 address of 10.162.107.246, subnet mask 255.255.255.0, and default gateway 10.162.107.195. The other two adapters show an IPv4 address of 192.168.127.1 and subnet mask 255.255.255.0. The prompt at the bottom shows 'PS C:\Users\Vaidehi> ipconfig' with the cursor on the next line.

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Wi-Fi 4:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :

Ethernet adapter VMware Network Adapter VMnet1:

Connection-specific DNS Suffix . :
IPv4 Address. . . . . : 192.168.127.1
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :

Ethernet adapter VMware Network Adapter VMnet8:

Connection-specific DNS Suffix . :
IPv4 Address. . . . . : 192.168.227.1
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
IPv4 Address. . . . . : 10.162.107.246
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 10.162.107.195
PS C:\Users\Vaidehi> ipconfig
```

Example Output:

Wireless LAN adapter Wi-Fi:

IPv4 Address. : 10.162.107.246

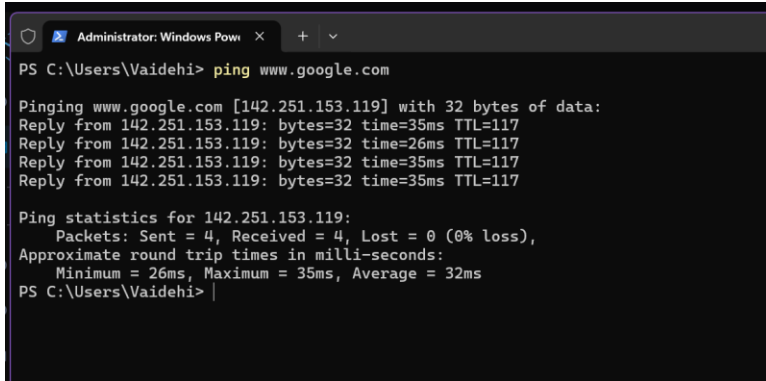
Subnet Mask : 255.255.255.0

Default Gateway : 10.162.107.195

Q.2 Use ping to check connectivity with Google

Solution: Test Connectivity Using Ping

I used the **ping** command to check whether my computer could communicate with Google's server.



```
Administrator: Windows Powe...
PS C:\Users\Vaidehi> ping www.google.com

Pinging www.google.com [142.251.153.119] with 32 bytes of data:
Reply from 142.251.153.119: bytes=32 time=35ms TTL=117
Reply from 142.251.153.119: bytes=32 time=26ms TTL=117
Reply from 142.251.153.119: bytes=32 time=35ms TTL=117
Reply from 142.251.153.119: bytes=32 time=35ms TTL=117

Ping statistics for 142.251.153.119:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 26ms, Maximum = 35ms, Average = 32ms
PS C:\Users\Vaidehi>
```

Example Output:

Pinging www.google.com [142.250.183.196] with 32 bytes of data:

Reply from 142.250.183.196: bytes=32 time=23ms TTL=117

Reply from 142.250.183.196: bytes=32 time=21ms TTL=117

Reply from 142.250.183.196: bytes=32 time=24ms TTL=117

Reply from 142.250.183.196: bytes=32 time=22ms TTL=117

Ping statistics for 142.250.183.196:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 26ms, Maximum = 35ms, Average = 32ms

Q.3 Use tracert to find the number of hops to Flipkart

Solution: Trace the Network Route

I used the **tracert** command to identify the path taken by packets from my computer to the Flipkart website.

```
Minimum = 26ms, Maximum = 35ms, Average = 32ms
PS C:\Users\Vaidehi> tracert www.flipkart.com

Tracing route to e348826.dsca.akamaiedge.net [23.55.244.178]
over a maximum of 30 hops:

  1  26 ms  12 ms  2 ms  10.162.107.195
  2  *      *      *      Request timed out.
  3  43 ms  38 ms  23 ms  192.168.197.209
  4  16 ms  18 ms  32 ms  nsg-corporate-34.208.187.122.airtel.in [122.187.208.34]
  5  133 ms  20 ms  26 ms  nsg-corporate-33.208.187.122.airtel.in [122.187.208.33]
  6  41 ms  55 ms  30 ms  116.119.36.180
  7  53 ms  106 ms  114 ms  ae8.gpx-bom2.netarch.akamai.com [23.55.247.248]
  8  123 ms  24 ms  38 ms  a23-55-244-178.deploy.static.akamaitechnologies.com [23.55.244.178]

Trace complete.
```

Result : Number of hops = 8

Q.4 Use nslookup to find Spotify's IP address

Solution: DNS Lookup Using nslookup

I used the **nslookup** command to find the IP address associated with www.spotify.com.

```
Trace complete.  
PS C:\Users\Vaidehi> nslookup www.spotify.com  
Server: UnKnown  
Address: 10.162.107.195  
  
Non-authoritative answer:  
Name: atc.spotify.map.fastly.net  
Addresses: 2a04:4e42:31::810  
151.101.211.42  
Aliases: www.spotify.com  
PS C:\Users\Vaidehi>
```

nslookup helps verify DNS resolution by checking which IP address is associated with a domain name.

Q.5 Troubleshoot a laptop that cannot access Zomato

Solution: Troubleshooting Sequence

If my friend cannot access Zomato, I would use the following three commands:

1. ipconfig

Purpose : I would first check whether the laptop has a valid **IPv4 address, subnet mask, and default gateway**.

For example:

IPv4 Address : 192.168.1.5

Default Gateway : 192.168.1.1

2. ping

First, I would test the default gateway : ping 192.168.1.1

Then I could test Internet connectivity : ping 8.8.8.8

Purpose : ping checks whether the computer can communicate with another network device or Internet host.

3. nslookup

nslookup www.zomato.com

Purpose : nslookup checks whether DNS can resolve www.zomato.com to an IP address.

If ping 8.8.8.8 works but nslookup www.zomato.com fails, the problem is likely related to **DNS resolution**.